

NCERT Exemplar Solutions for Class 12

Physics

Chapter 4 - Moving Charges and Magnetism

Exercise

MULTIPLE CHOICE QUESTIONS-I

4.1 Two charged particles traverse identical helical paths in a completely opposite sense in a uniform magnetic field $B = B_0 \hat{k}$.

- (a) They have equal z-components of momenta.**
- (b) They must have equal charges.**
- (c) They necessarily represent a particle-antiparticle pair.**
- (d) The charge to mass ratio satisfy: $\left(\frac{e}{m}\right)_1 + \left(\frac{e}{m}\right)_2 = 0$.**

Ans. (d) The charge to mass ratio satisfy: $\left(\frac{e}{m}\right)_1 + \left(\frac{e}{m}\right)_2 = 0$.

For a given pitch, $P = \frac{2\pi m v \cos \theta}{Bq}$,

$$\frac{q}{m} = \frac{2\pi v \cos \theta}{BP} \quad [\text{Where } \theta \text{ is the angle of velocity of charge particle with x - axis}]$$

If motion is not helical then $\theta = 0$.

As the path of both the particles is identical and helical but of opposite direction in same magnetic field so, by the law of conservation of momenta.

$$\left(\frac{e}{m}\right)_1 + \left(\frac{e}{m}\right)_2 = 0$$

So, verifies Ans (d).

4.2 Biot-Savart law indicates that the moving electrons (velocity v) produce a magnetic field B such that

- (a) $B \perp v$**
- (b) $B \parallel v$**
- (c) It obeys inverse cube law.**
- (d) It is along the line joining the electron and point of observation.**

Ans. (a) $B \perp v$

By Biot-Savart law,

$$dB = \frac{I \cdot dl \sin \theta}{r^2}$$

$$\text{Or } dB = \frac{I \times dl}{r}$$

According to Biot-Savart law, if the magnetic field is not perpendicular to the motion of charge then it will not move in a helical path, which is not possible for motion of a charge in magnetic field.

So the magnetic field is perpendicular to the direction of flow of charge verifies Ans 'a'.

4.3 A current carrying circular loop of radius R is placed in the $x - y$ plane with center at the origin. Half of the loop with $x > 0$ is now bent so that it now lies in the $y - z$ plane.

- (a) The magnitude of magnetic moment now diminishes.**
- (b) The magnetic moment does not change.**
- (c) The magnitude of B at $(0, 0, z)$, $z \gg R$ increases.**
- (d) The magnitude of B at $(0, 0, z)$, $z \gg R$ is unchanged.**

Ans. (a) the magnitude of magnetic moment now diminishes.

According to Fleming's left hand rule the direction of magnetic field due to a current-carrying circular loop is perpendicular to the plane of loop and unidirectional.

In first case, the direction of magnetic field is only in positive $x - z$ direction but when it is bent then magnetic field due to first half loop is along $-x$ axis (due to unfolded

loop) and second half loop is along +x axis direction. It is equal in magnitude so vector sum of equal and opposite magnetic field will cancel out each other.

Hence, this verifies Ans (a).

4.4 An electron is projected with uniform velocity along the axis of a current carrying long solenoid. Which of the following is true?

- (a) The electron will be accelerated along the axis.**
- (b) The electron path will be circular about the axis.**
- (c) The electron will experience a force at 45° to the axis and hence execute a helical path.**
- (d) The electron will continue to move with uniform velocity along the axis of the solenoid.**

Ans. (d) The electron will continue to move with uniform velocity along the axis of the solenoid.

The Lorentz force acts on a charged particle in a magnetic and electric field is

$$F = q(\mathbf{v} \times \mathbf{B})$$

$$F = qvB\sin\theta$$

As the charge is moving in the direction of electric field, $\theta = 0$. The force due to the electric field will also be zero, so it will not affect the velocity of moving charge particle.

Hence, this verifies Ans (d).

4.5 In a cyclotron, a charged particle

- (a) Undergoes acceleration all the time.**
- (b) Speeds up between the dees because of the magnetic field.**
- (c) Speeds up in a dee.**
- (d) Slows down within a dee and speeds up between dees.**

Ans. (b) Speeds up between the dees because of the magnetic field.

The electric field and magnetic field are perpendicular to each other. When a charge is placed between dees in empty space then it will experience a force due to electric field and magnetic field which keeps it into a circular path.

Inside the dees, there is no electric field due to shielding effect of charge so the magnetic field is only responsible for the circular motion of charged particle. Inside dees the charge particle will move with constant velocity.

So the charged particle accelerates inside dees and hence, this verifies Ans (b).

4.6 A circular current loop of magnetic moment M is in an arbitrary orientation in an external magnetic field B . The work done to rotate the loop by 30° about an axis perpendicular to its plane is

(a) MB .

(b) $\frac{\sqrt{3}}{2}MB$

(c) $\frac{MB}{2}$

(d) Zero.

Ans. (b) $\frac{\sqrt{3}}{2}MB$ and (d) Zero

When the axis of rotation of a loop is along B , then the angle between $\vec{B}$ and $\vec{A}$ is 90° always. So WD by loop to rotate i.e., $WD = MB \cos 90^\circ$ which is zero.

This verifies the option (d).

But, when the axis of rotation of the loop is not along the direction of B , then the direction of vector $\vec{B}$ and $\vec{A}$ will change with time.

Work done by the loop during the orientation in uniform magnetic field is

$$= MB(\cos \theta_2 - \cos \theta_1) = MB \cos \theta$$

$$= MB \cos 30^\circ = MB \frac{\sqrt{3}}{2}$$

Hence, the Ans (b) is verified.

MULTIPLE CHOICE QUESTIONS-II MORE THAN ONE OPTION

4.7 The gyro-magnetic ratio of an electron in an H-atom, according to Bohr's model, is

- (a) Independent of which orbit it is in.
- (b) Negative.
- (c) Positive.
- (d) Increases with the quantum number n .

Ans.(b) Negative

Angular momentum of electron is $L = \frac{nh}{2\pi}$

Magnetic moment of electron is $M = n \left[\frac{eh}{4\pi m} \right]$

Gyromagnetic ratio = $\frac{\text{magnetic moment}}{\text{angular momentum}}$

Gyromagnetic ratio = $\frac{2\pi neh}{4\pi nhm}$

Gyromagnetic ratio = $\frac{e}{2m}$

Gyromagnetic ratio = constant

It is independent of the orbit in which electron is revolving.

Since e is negative, the gyromagnetic ratio is negative.

Hence, this verifies the Ans is (b)

4.8 Consider a wire carrying a steady current I placed in a uniform magnetic field B perpendicular to its length. Consider the charges inside the wire. It is known that magnetic forces do no work. This implies that,

- (a) Motion of charges inside the conductor is unaffected by B since they do not absorb energy.
- (b) Some charges inside the wire move to the surface as a result of B .

(c) If the wire moves under the influence of B, no work is done by the force.

(d) If the wire moves under the influence of B, no work is done by the magnetic force on the ions, assumed fixed within the wire.

Ans. (b) Some charges inside the wire move to the surface as a result of B and (d) if the wire moves under the influence of B, no work is done by the magnetic force on the ions, assumed fixed within the wire.

Force F on current-carrying conductor by magnetic field B is given by the formula

$$F = BIL \sin \theta = I(B \times L)$$

Since the wire is placed perpendicular to the magnetic field B, so the angle between normal to the wire and magnetic field is 0° .

$$\sin 0^\circ = 0$$

$$\text{So } F = 0$$

As there is no force on charges so their motion will remain unaffected.

This verifies the Ans (b).

So the work done by magnetic field is $W = F \cdot I \cos \theta$

$$W = F \cdot I \cos 90^\circ \text{ which is equal to zero.}$$

Hence, this verifies the Ans (d).

4.9 Two identical current carrying coaxial loops carry current I in an opposite sense. A simple amperian loop passes through both of them once. Calling the loop as C,

(a) $\oint_C \vec{B} \cdot d\vec{l} = \mp 2\mu_0 I$

(b) The value of $\oint_C \vec{B} \cdot d\vec{l}$ is independent of sense of C.

(c) There may be a point on C where $\vec{B}$ and $d\vec{l}$ are perpendicular.

(d) B vanishes everywhere on C.

Ans. (b) the value of is independent of sense of C and (c) There may be a point on C where and are perpendicular.

Loops are identical and placed coaxially with carrying same current in opposite sense. So, inside the amperian loop of any type the direction of current will be opposite according to Ampere's circuital law

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0(I - I) = \mu_0(0) = 0$$

As the magnetic field inside (over everywhere) the loop is perpendicular to the direction of plane of loop, so

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = |\mathbf{B} \cdot d\mathbf{l}| \cos 90^\circ = 0$$

So, Ans (b) and (c) are verified.

4.10 A cubical region of space is filled with some uniform electric and magnetic fields. An electron enters the cube across one of its faces with velocity v , and a positron enters via opposite face with velocity $-v$. At this instant,

- (a) The electric forces on both the particles cause identical accelerations.**
- (b) The magnetic forces on both the particles cause equal accelerations.**
- (c) Both particles gain or lose energy at the same rate.**
- (d) The motion of the center of mass (CM) is determined by 'B' alone.**

Ans. (b) The magnetic forces on both the particles cause equal accelerations, (c) Both particles gain or lose energy at the same rate and (d) The motion of the center of mass (CM) is determined by 'B' alone.

As the $F = qE$ here E is same but q is in opposite nature force. Electric force or acceleration is not identical. It discards Ans (a).

As the $F = q(\mathbf{v} \times \mathbf{B})$ i.e. F is perpendicular to velocity and magnetic field, so particle revolves perpendicular to both and with uniform speed. But magnitude of acceleration by magnetic field is equal. It verifies Ans (b).

As magnitude of charge $\vec{v}$, $\vec{E}$ and B are constant, so gain or lose the energy at the same rate verifies Ans (c).

As there is no change in center of mass of particles therefore the motion of center of mass is determined by B alone. It verifies Ans (d).

4.11 A charged particle would continue to move with a constant velocity in a region wherein,

(a) $E = 0, B \neq 0$,

(b) $E \neq 0, B \neq 0$,

(c) $E \neq 0, B = 0$,

(d) $E = 0, B = 0$

Ans. (a) $E = 0, B \neq 0$, (b) $E \neq 0, B \neq 0$ and (d) $E = 0, B = 0$

A charged particle will move with a constant velocity in a region if the force on it due to both the electric field and magnetic field is zero, or force on the charged particle due to electric field is equal and opposite to the force on it due to magnetic field.

i.e. $F_e = qE$ and

$$F_m = q(v \times B) = qvB \sin \theta$$

$F_e = 0$ if $E = 0$ and $F_m = 0$ if $\theta = 0$ or 180

So, $F_e = 0$ if $E = 0$ and the particle is moving in the direction of magnetic field i.e. $B \neq 0$ or $E = 0; B = 0$

The resultant force $-qE + q(v \times B) = 0$

i.e. $E \neq 0, B \neq 0$

VERY SHORT ANSWER TYPE QUESTIONS

4.12 Verify that the cyclotron frequency $\omega = \frac{eB}{m}$ has the correct dimensions of $[T]^{-1}$.

Ans. In cyclotron, charged particles are placed perpendicular to both E and B . Electric field exerts force on particle resulting increase in velocity of particle and magnetic field keeps it into circular path. So it provides the centripetal force for revolution. So,

$$\frac{mv^2}{R} = qv \times B.$$

Here, $\theta = 90^\circ$ as θ is angle between $\vec{v}$ and $\vec{B}$

$$\therefore \frac{mv^2}{R} = qvB \Rightarrow \frac{v^2}{Rv} = \frac{qB}{m} \text{ or } \frac{qB}{m} = \frac{v}{R}$$

$\therefore w = \frac{qB}{m}$ so dimensions of below must be equal.

$$\text{So } [\omega] = \left[\frac{qB}{m} \right] = \left[\frac{v}{R} \right]$$

$$\left[\frac{2\pi}{T} \right] = \left[\frac{LT^{-1}}{L} \right] = [T^{-1}]$$

So dimensions of ω is $[T]^{-1}$.

4.13 Show that a force that does no work must be a velocity dependent force.

Ans. The work done by a force is given by $dW = F \cdot dl = Fdl \cos \theta$.

As work done by force is zero, so

$$dW = F \cdot dl = 0$$

$$\Rightarrow F \cdot \frac{dl}{dt} \times dt = 0$$

$$dW = F \cdot v dt = 0$$

$$\text{as } dt \neq 0 [\therefore F \cdot v = 0]$$

So F must be velocity-dependent, i.e. angle between F and v must be always, then

$$\text{For } F \cdot v = 0$$

$$Fv \cos \theta = \cos 90^\circ$$

$$\theta = 90^\circ$$

If v changes direction then to make $\theta = 90^\circ$, F must change angle according to v . So F is dependent of v to make work done zero.

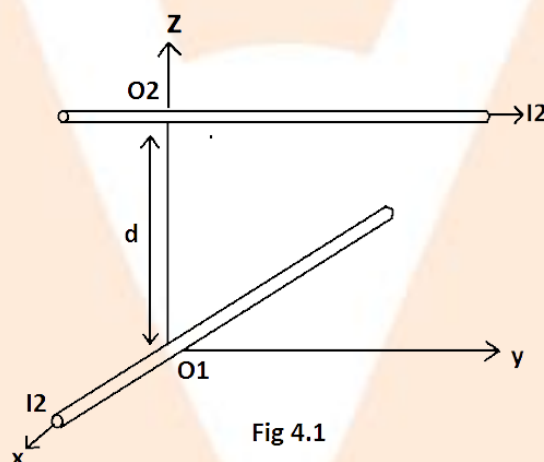
4.14 The magnetic force depends on velocity vector which depends on the inertial frame of reference. Does then the magnetic force differ from inertial frame to frame? Is it reasonable that the net acceleration has a different value in different frames of reference?

Ans. The magnetic force changes from inertial frame to frame, i.e. the magnetic force depends on frame of reference. So the net acceleration which comes into existence out of this is, however, frame independent (non-relativistic physics) for inertial frame.

4.15 Describe the motion of a charged particle in a cyclotron if the frequency of the radio frequency (rf) field were doubled.

Ans. When the frequency ω of electric field (oscillator) is doubled, the time-period $\left[T = \frac{2\pi}{\omega}\right]$ becomes half. So the charged particle will take half time to reach between dees. Hence, a charged particle accelerates as it moves in circular path between the dees during motion in Dee's the radius of moving charged particle remain same.

4.16 Two long wires carrying current I_1 and I_2 are arranged as shown in Fig. 4.1. The one carrying current I_1 is along is the x-axis. The other carrying current I_2 is along a line parallel to the y-axis given by $x = 0$ and $z = d$. Find the force exerted at O_2 because of the wire along the x-axis.



Ans. We know that the force on current (I) carrying conductor placed in magnetic field B is

$$F = I(L \times B) = ILB\sin\theta$$

The direction of magnetic field at O_2 due to the current I_1 is parallel to the Y – axis and in $-Y$ direction.

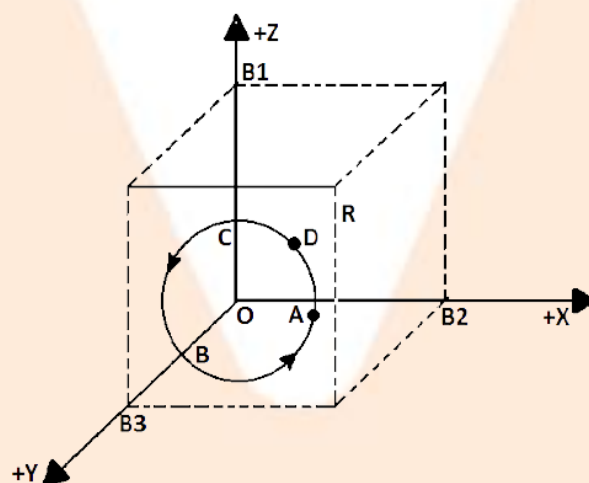
As wire of current I_2 is parallel to Y – axis, current in I_2 is also along Y – axis. So I_2 and B_1 (magnetic field due to current I_1) are also along Y – axis i.e. angle between I_2 and B_1 is zero. So magnetic force F_2 on wire of current I_2 is $F_2 = B_1 I_2 L_1 \sin 0^\circ = 0$

Hence, force on O_2 due to wire of current I_1 is zero.

SHORT ANSWER TYPE QUESTIONS

4.17 A current carrying loop consists of 3 identical quarter circles of radius R , lying in the positive quadrants of the X – Y , Y – Z and Z – X planes with their centers at the origin, joined together. Find the direction and magnitude of B at the origin.

Ans. In the figure, consider the 3 quadrants of conductors AB , BC and CD along positive X – Y , Y – Z and Z – X planes respectively. A and D are connected to a battery which is responsible for the flow of current I through the three quadrants of radius R in the coordinate of A or D ($R, 0, 0$), $B(0, R, 0)$ and of $C(0, 0, R)$. Now the direction of magnetic field by right-hand thumb rule due to quadrants AB , BC and CD are $+B_1$, B_2 and B_3 along $+Z$, $+X$ and $+Y$ directions respectively. So, at the center of quadrant



$$\therefore \vec{B} = \frac{\mu_0 I}{2\pi R} \cdot \frac{\pi}{2}$$

$$B = \frac{\mu_0 I}{8\pi R}$$

So M.F. due to quadrants AB, BC and CD at their center O are B_1 , B_2 and B_3 respectively.

$$B_1 = \frac{\mu_0 I}{8\pi R} \hat{k}, B_2 = \frac{\mu_0 I}{8\pi R} \hat{i} \text{ and } B_3 = \frac{\mu_0 I}{8\pi R} \hat{j}.$$

So net magnetic field at origin due to three current carrying loops are $B = B_1 + B_2 + B_3$.

$$B = \frac{\mu_0 I}{8\pi R} [\hat{i} + \hat{j} + \hat{k}]$$

The resultant of B_1 , B_2 and B_3 will be diagonal OR of cube of side B_1 , B_2 , B_3 as the $|B_1| = |B_2| = |B_3|$.

4.18. A charged particle of charge e and mass m is moving in an electric field $\vec{E}$ and magnetic field $\vec{B}$. Construct dimensionless quantities and quantities of dimension $[T]^{-1}$.

Ans. When a charged particle is placed in an electric and magnetic field, it starts moving in a helical motion where electric field E and B is perpendicular to direction of motion and electric field exerts a force on charged particle and centripetal force is applied by magnetic force

$$F_m = qvB \sin 90^\circ = qvB \dots (1)$$

We know that Centripetal force is $\frac{mv^2}{R} \dots (2)$

By equation (1) and (2)

$$\frac{mv^2}{R} = qvB$$

$$\frac{v}{R} = \frac{qB}{m}$$

$$\therefore v = \omega R \text{ and } q = e$$

Angular velocity

$$\omega = \frac{v}{R} = \frac{eB}{m}$$

Dimensional formula for angular velocity ω

$$\omega = \left[\frac{eB}{m} \right] = \left[\frac{v}{R} \right] = [T^{-1}]$$

4.19 An electron enters with a velocity $v = v_0 \hat{i}$ into a cubical region (faces parallel to coordinate planes) in which there are uniform electric and magnetic fields. The orbit of the electron is found to spiral down inside the cube in plane parallel to the X – Y plane. Suggest a configuration of fields $\vec{E}$ and $\vec{B}$ that can lead to it.

Ans. The velocity of electron is $v = v_0 \hat{i}$, i.e. along X – axis as magnetic field is perpendicular to velocity so it is in Y direction.

The moving electron enters into cubical region. The force on electron due to Lorentz force

$$F = q(E + v \times B)$$

By putting the values

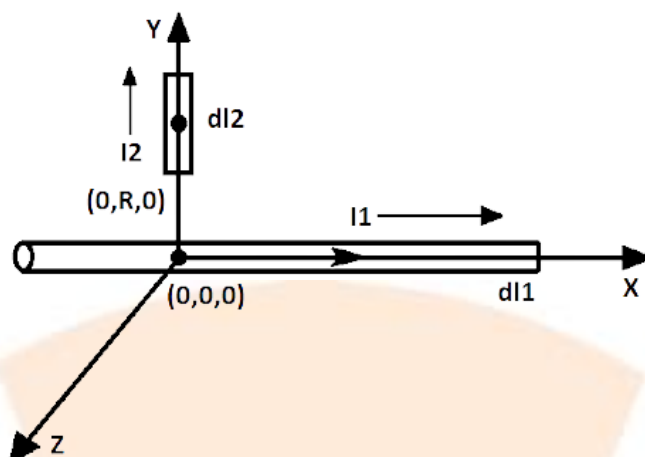
$$F_m = -e[v_0 \hat{i} \times B \hat{j}] = -ev_0 B \hat{k}$$

Which revolve the electron in X – Y plane.

The force due to electric field $F_e = e\vec{E}K$ accelerates the electron along z – axis and force due to magnetic field keeps it in circular motion, which in turn increases the radius of circular path. So the motion becomes helical path.

4.20 Do magnetic forces obey Newton's third law? Verify for two current elements $dl_1 = dl \hat{i}$ located at the origin and $dl_2 = dl \hat{j}$ located at $(0, R, 0)$. Both carry current I.

Ans. Fig. shows two current elements dl_1 and dl_2 located at $(0, 0, 0)$ and $(0, R, 0)$. The direction of magnetic field on dl_2 due to magnetic field B_1 by dl_1 will be along +Z direction by Right Hand Grip Rule i.e. directions of I_2 and B_1 are perpendicular.



Force on dl_2 due to dl_1

$$F_1 = B_1 I_2 dl_2 \sin 90^\circ = B_1 I_2 dl_2$$

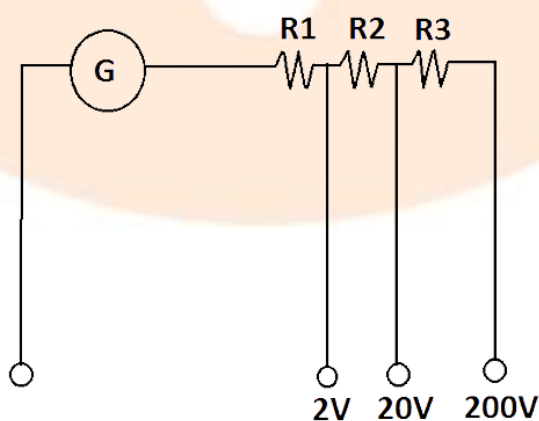
Similarly, angle between magnetic field B_2 and current I_1 is 0° . So the force acting on dl_1 due to dl_2

$$F_2 = B_2 I_1 dl_1 \sin 0^\circ = 0$$

$$\therefore |dl_1| = |dl_2| = |dl|$$

So the magnetic force existing on dl_1 is zero but on dl_2 due to dl_1 is not zero so magnetic forces do not obey Newton's third law.

4.21 A multirange voltmeter can be constructed by using a galvanometer circuit, as shown in Fig. 4.2. We want to construct a voltmeter that can measure 2V, 20V and 200V using a galvanometer of resistance 10Ω and that produces maximum deflection for current of 1mA. Find R_1 , R_2 and R_3 that have to be used.



Ans. Given resistance of Galvanometer (R_g) is 10Ω .

Maximum current deflection (I_g) is $1\text{mA} = 1 \times 10^{-3}\text{A}$

For 2V , R_g and R_1 are connected in series:

$$I_g(R_g + R_1) = 2\text{V}$$

$$1 \times 10^{-3} [10 + R_1] = 2$$

$$R_1 = 2000 - 10 = 1990\Omega$$

For 20V , R_g , R_1 and R_2 are connected in series:

$$I(R_g + R) = V \quad \because R = R_1 + R_2$$

$$I_g [R_g + R_1 + R_2] = 20\text{V}$$

$$10^{-3} [10 + 1990 + R_2] = 20$$

$$R_2 = 20000 - 2000$$

$$R_2 = 18000\Omega = 18\text{k}\Omega$$

For 200V , R_g , R_1 , R_2 and R_3 are connected in series:

$$I_g [R_g + R_1 + R_2 + R_3] = 200\text{V}, \text{ as } R = (R_1 + R_2) + R_3$$

$$10^{-3} [10 + 1990 + 18000 + R_3] = 200$$

$$20000 + R_3 = 200000$$

$$R_3 = 200000 - 20000$$

$$= 180000\Omega = 180\text{k}\Omega$$

Hence, $R_1 = 1990\Omega$; $R_2 = 18\text{k}\Omega$; $R_3 = 180\text{k}\Omega$

4.22 A long straight wire carrying current of 25A rests on a table as shown in Fig. **4.3**. Another wire PQ of length 1m , mass 2.5g carries the same current but in the opposite direction. The wire PQ is free to slide up and down. To what height will PQ rise?

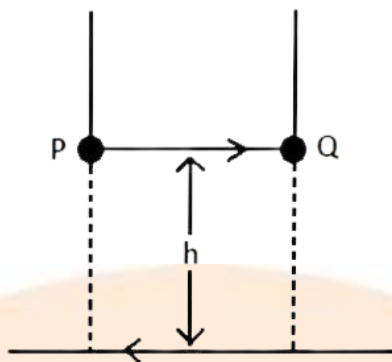


Fig. 4.3

Ans. Given current in wire (I_2) = 25 A ,

Length of wire PQ (L_2) = 1 m ,

Mass of wire PQ (m) = 2.5g

Wire PQ must experience a repulsive force due to magnetic field by wire AB .

Let the wire is balanced at height h thus, magnetic force due to wire AB on PQ should be equal and opposite to weight of the body ($= mg$).

$$F_m = mg \dots\dots (1)$$

Also the magnetic force due to a wire

$$F_m = ILB \sin \theta \dots\dots\dots (2)$$

The angle between B_1 and I in PQ is 90° . So by equation (1) and (2) we get,

$$ILB = mg \left(\sin 90^\circ = 1 \right)$$

$$B_1 I_2 L_2 = mg$$

$$\frac{\mu_0 I_1}{2\pi h} I_2 l_2 \sin 90^\circ = mg \quad [I_1 = I_2 = I = 25A]$$

$$\frac{\mu_0 I^2 l_2}{2\pi h} = mg$$

$$h = \frac{\mu_0 I^2 l_2}{2\pi mg} = \frac{4\pi \times 10^{-7} \times 25 \times 25 \times 1}{2\pi \times 2.5 \times 10^{-3} \times 9.8}$$

$$= \frac{2 \times 625}{2.5 \times 9.8} \times 10^{-7+3}$$

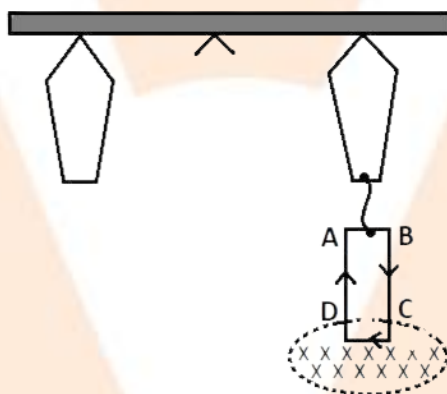
$$= \frac{20 \times 25}{9.8} \times 10^{-4}$$

$$= 51 \times 10^{-4} \text{ m}$$

$$= 51 \times 10^{-2} \text{ cm} = 0.51 \text{ cm}$$

LONG ANSWER TYPE QUESTIONS

4.23 A 100 turn rectangular coil ABCD (in XY plane) is hung from one arm of a balance (Fig. 4.4). A mass 500g is added to the other arm to balance the weight of the coil. A current 4.9A passes through the coil and a constant magnetic field of 0.2T acting inward (in xz plane) is switched on such that only arm CD of length 1cm lies in the field. How much additional mass 'm' must be added to regain the balance?



Ans. The magnetic field is perpendicular to arms BC and AD, so torque will act on CD and AB arms due to it, Coil rotate.

When current of 4.9 A does not pass through the coil the balance measures mass of coil 500 g.

On arm AD and BC of rectangular coil magnetic force due to M.F. will be equal and opposite so it will rotate the coil horizontally not vertically up or down. So does not affect the balance.

When current of 4.9 A passes through the coil, downward force acts on arm CD due to magnetic field. Length of arm CD is 1 cm (10^{-2} m).

Force acting on arm CD = $F_m = B \times I \cdot l = BIl \sin \theta$ [θ is angle between B and I in CD]

$$= 0.2 \times 4.9 \times \sin 90^\circ \times 10^{-2}$$

$$\text{Or } F_m = 0.98 \times 10^{-2} \text{ N}$$

Now let the weight mg is added on other side of beam balance to balance the coil

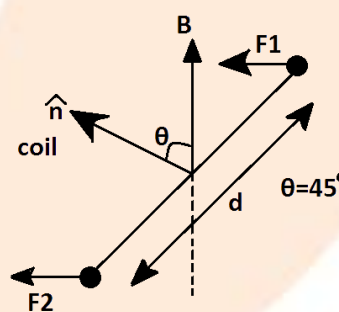
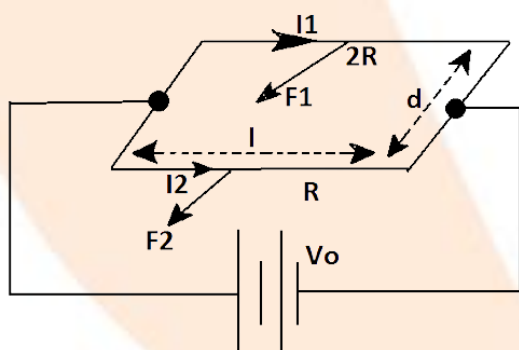
$$Mg = F_m$$

$$m \times 9.8 = 0.98 \times 10^{-2}$$

$$m = \frac{0.98}{9.8} \times 10^{-2} = 10^{-3} \text{ kg} = 1 \text{ g}$$

4.24 A rectangular conducting loop consists of two wires on two opposite sides of length 'l' joined together by rods of length 'd'. The wires are each of the same material but with cross-sections differing by a factor of 2. The thicker wire has a resistance R and the rods are of low resistance, which in turn are connected to a constant voltage source V_0 . The loop is placed in uniform a magnetic field B at 45° to its plane. Find (τ), the torque exerted by the magnetic field on the loop about an axis through the centers of rods.

Ans.



Force on axis AB due to magnetic field is $F_1 = I_1 l B$

$$= \frac{V_0}{R_1} l B = \frac{V_0}{2R} l B$$

It is acting perpendicular to the arm AB towards observe.

Torque due to force F_1 about axis EF

$$\tau_1 = F_1 \times \frac{d}{2} \sin 45^\circ$$

$$\Rightarrow \tau_1 = \frac{V_0}{2R} lB \times \frac{d}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{V_0 lBd}{4\sqrt{2}R}$$

(In the anticlockwise direction)

Now the force on arm DC due to magnetic field,

$$F_2 = I_2 lB$$

$$\Rightarrow F_2 = \frac{V_0}{R_2} lB$$

$$\Rightarrow F_2 = \frac{V_0}{R} lB$$

Torque due to force F_2 about axis EF is

$$\tau_2 = F_2 \times \frac{d}{2} \sin 45^\circ$$

$$\tau_2 = \frac{V_0}{R} lB \times \frac{d}{2} \times \frac{1}{\sqrt{2}}$$

$$\tau_2 = \frac{V_0 lBd}{2\sqrt{2}R}$$

(In the clockwise direction)

Net Torque is $\tau = \tau_2 - \tau_1$

$$\Rightarrow \frac{V_0 lBd}{2\sqrt{2}R} - \frac{V_0 lBd}{4\sqrt{2}R}$$

$$\Rightarrow \frac{V_0 lBd}{2\sqrt{2}R} \left(1 - \frac{1}{2} \right)$$

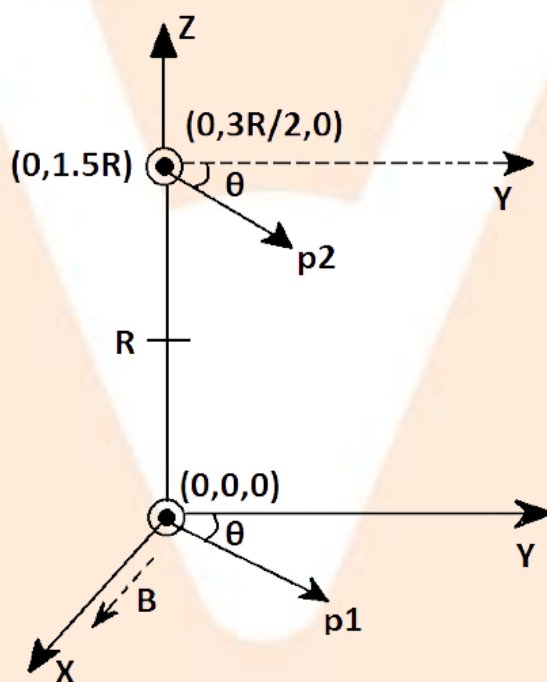
$$\Rightarrow \frac{V_0 lBd}{4\sqrt{2}R}$$

But, $l \times d = A$

$$\text{So, } \tau = \frac{V_0 l B d}{4\sqrt{2}R}$$

4.25 An electron and a positron are released from $(0,0,0)$ and $(0,0,1.5R)$ respectively in a uniform magnetic field $B = B_0 \hat{i}$, each with an equal momentum of magnitude $\vec{p} = e\vec{B}R$. Under what conditions on the direction of momentum will the orbits be non-intersecting circles?

Ans. As $B = B_0 \hat{i}$ so magnetic field is along $+X$ -axis. The circular motion of the momenta of both an electron and a positron are in $Y-Z$ plane. Let p_1 and p_2 are the momentum of the electron and positron respectively, the magnitude of charge and momentum of both are equal so they revolve in $Y-Z$ plane due to MF is $B = B_0 \hat{i}$. But in opposite sense with same radius R as the direction, p_1 and p_2 are opposite.



Let p_1 and p_2 make an angle $+\theta$ and $-\theta$ with Y -axis, as shown in the above figure. The centers of the respective circles must be perpendicular to the momentum and at a distance R . Let the Centre of revolving electron and positron are C_e and C_p respectively so coordinates of C_e will be $(0, +R \sin\theta, R \cos\theta)$ and that of C_p will be $(0, -R \sin\theta, \frac{3}{2}R - R \cos\theta)$

The planes of circular paths are in $Y - Z$ plane.

The condition that the circular path of the electron and positron are non-intersecting circles is that the distance (d) between their centers must be greater than $2R$

$$d > 2R$$

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$$

$$d^2 = (\mathbf{0} - \mathbf{0})^2 + (+R \sin \theta + R \sin \theta)^2 + \left(R \cos \theta - \frac{3}{2}R + R \cos \theta \right)^2$$

$$d^2 = (2R \sin \theta)^2 + \left(2R \cos \theta - \frac{3}{2}R \right)^2$$

$$d^2 = 4R^2 \sin^2 \theta + 4R^2 \cos^2 \theta + \frac{9}{4}R^2 - 6R^2 \cos \theta$$

$$d^2 = 4R^2 + \frac{9}{4}R^2 - 6R^2 \cos \theta \dots (I)$$

$$d > 2R$$

$$d^2 > 4R^2$$

$$4R^2 + \frac{9}{4}R^2 - 6R^2 \cos \theta > 4R^2$$

$$\frac{9}{4}R^2 - 6R^2 \cos \theta > 0$$

$$3R^2 \left(\frac{3}{4} - 2 \cos \theta \right) > 0$$

$$3R^2 > 0 \text{ is rejected}$$

$$2 \cos \theta < \frac{3}{4}$$

$\cos \theta < \frac{3}{8}$ is the condition that two circular paths do not intersect; θ is angle of momentum of electron or positron with Y - axis.

4.26 A uniform conducting wire of length $12a$ and resistance R is wound up as a current carrying coil in the shape of (i) an equilateral triangle of side a ; (ii) a square of sides a and, (iii) a regular hexagon of sides a . The coil is connected to a voltage source V_0 . Find the magnetic moment of the coils in each case.

Ans. (i) The coil of shape equilateral triangle is made up with side a . So the number of turns in coil $= \frac{12a}{3a} = 4$ turns.

$$\therefore \text{Magnetic moment} = nIA = 4 \times I \times \frac{\sqrt{3}}{4} a^2$$

Magnetic moment of the triangular coil $= \sqrt{3} Ia^2$.

(ii) Number of turns in coil of square-shape of side $a = \frac{12a}{4a} = 3$ turns. So magnetic moment due to square shaped coil $= nIA = 3I \times a^2 = 3Ia^2$

(iii) For a regular hexagon-shaped coil of side a , number of turns in coil $= \frac{12a}{6a} = 2$

$$\therefore \text{Magnetic moment} = nIA = 2I \cdot \left(\frac{\sqrt{3}}{4} a^2 \right) \cdot 6 = 3\sqrt{3} Ia^2$$

4.27. Consider a circular current-carrying loop of radius R in the $x-y$ plane with center at origin. Consider the line integral

$$\mathfrak{Z}(L) = \left| \int_{-L}^L \mathbf{B} \cdot d\mathbf{l} \right| \text{ taken along } Z\text{-axis}.$$

(a) Show that $\mathfrak{Z}(L)$ monotonically increases with L .

(b) Use an appropriate Amperian loop to show that $\mathfrak{Z}(\infty) = \mu_0 I$, where I is the current in the wire.

(c) Verify directly the above result.

(d) Suppose we replace the circular coil by a square coil of sides R carrying the same current I . What can you say about $\mathfrak{Z}(L)$ and $\mathfrak{Z}(\infty)$?

Ans. (a) $B(z)$ point in the same direction of Z -axis and hence $\oint B \cdot dl$ is monotonical function of L as B and dl are along the same direction. So

$$B \cdot dl = B dl \cos \theta = B dl \cos 0^\circ = B dl$$

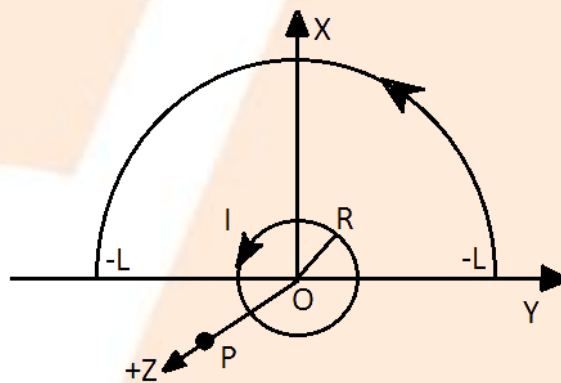
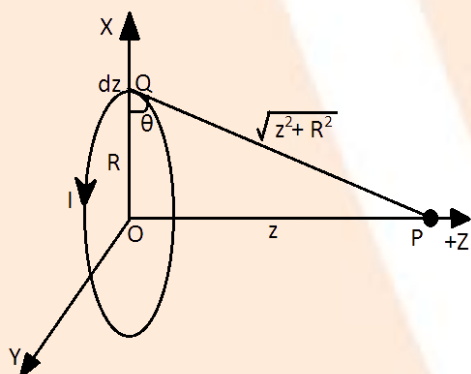
(b) $\oint B \cdot dl$ + Contribution from large distance on contour $C = \mu_0 I$

$$\therefore \text{As } L \rightarrow \infty$$

$$\text{Contribution from large distance} \rightarrow 0 \quad \left(\text{As } B \propto \frac{1}{r^3} \right)$$

$$\therefore \oint B \cdot dl = \mu_0 I$$

(c) The magnetic field due to circular current-carrying loop of radius R in $X-Y$ plane with center at origin at any point lying at distance a from origin. Consider a loop of current-carrying conductor placed in $X-Y$ plane. A point P is in $+Z$ direction at distance z , i.e. $OP = z$.



Again consider an element dz on loop of conductor as shown in figure below.

Let angle between R and $QP = \theta$ then magnetic field at P due to loop is

$$B_z = \frac{\mu_0 I R^2}{2(z^2 + R^2)^{3/2}} \quad [\text{Integrating both sides w.r.t. } z \text{ and } z \text{ can vary from } -d \text{ to } +d]$$

$$\int_{-\infty}^{\infty} B_z dz = \int_{-\infty}^{\infty} \frac{\mu_0 I R^2 dz}{2(z^2 + R^2)^{3/2}}$$

$$\tan \theta = \frac{z}{R}$$

$$z = R \tan \theta$$

Differentiating both sides:

$$dz = R \cdot \sec^2 \theta \cdot d\theta$$

$$\cos \theta = \frac{R}{\sqrt{z^2 + R^2}}$$

$$\cos^2 \theta = \frac{R}{\sqrt{z^2 + R^2}}$$

$$\int_{-\infty}^{\infty} B_z dz = \frac{\mu_0 I}{2} \int_{-\infty}^{+\infty} \frac{R^2}{(z^2 + R^2)} \frac{dz}{\sqrt{z^2 + R^2}}$$

$$\int_{-\infty}^{\infty} B_z dz = \frac{\mu_0 I}{2} \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} \cos^2 \theta \cdot \frac{R \sec^2 \theta \cdot d\theta}{\sqrt{z^2 + R^2}} \left[\because \cos \theta = \frac{R}{\sqrt{z^2 + R^2}} \right]$$

$$\int_{-\infty}^{\infty} B_z dz = \frac{\mu_0 I}{2} \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} \frac{R d\theta}{\sqrt{z^2 + R^2}} = \frac{\mu_0 I}{2} \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} \cos \theta d\theta$$

$$\int_{-\infty}^{\infty} B_z dz = \frac{\mu_0 I}{2} [\sin \theta]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{\mu_0 I}{2} \left[\sin \frac{\pi}{2} - \sin \frac{-\pi}{2} \right]$$

$$\int_{-\infty}^{\infty} B_z dz = \frac{\mu_0 I}{2} [1 + 1] = \mu_0 I = \int_{-\infty}^{+\infty} \hat{B}_z dz = \mu_0 I$$

(d) Because the area of square loop is smaller than the area of circular loop, for the same length of conducting wires, hence $\text{loop } B(z)_{\text{square loop}} < \text{loop } B(z)_{\text{circular loop}}$

$$\mathfrak{I}_S(L)_{\text{sq.}} = \mathfrak{I}_S(L)_{\text{circular}} \quad [\because \text{Length of conducting wire is equal}]$$

By using arguments as in (b) part, B_z does not depend on length of wire

$$\therefore \mathfrak{I}_S(\infty)_{\text{sq. loop}} = \mathfrak{I}_S(\infty)_{\text{circular loop}}$$

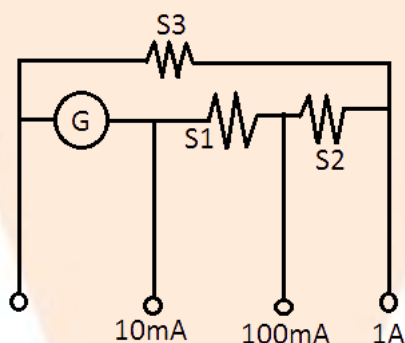
Magnetic field due to circular or square loops remains same, i.e.

$$(B_z)_{\text{circular loop}} = (B_z)_{\text{square loop}} = \mu_0 I$$

$$\text{So, } \mathfrak{I}(\infty)_{\text{square loop}} = \mathfrak{I}(\infty)_{\text{circular loop}} = \mu_0 I$$

Hence proved.

4.28 A multi-range current meter can be constructed by using a galvanometer circuit as shown in Fig. 4.5. We want a current meter that can measure 10mA, 100mA and 1A using a galvanometer of resistance 10Ω and that produces maximum deflection for current of 1mA. Find S_1 , S_2 and S_3 that have to be used.



Ans. We can measure the currents of magnitude 10mA, 100mA and 1A by connecting ammeter A and B, C and D respectively. So,

$$\text{For } 10\text{mA} \rightarrow I_g G = (I - I_g)(S_1 + S_2 + S_3) \dots\dots\dots (I)$$

$$\text{For } 100\text{mA} \rightarrow I_g (G + S_1) = (I - I_g)(S_2 + S_3) \dots\dots\dots (II)$$

$$\text{For } 1\text{A} \rightarrow I_g (G + S_1 + S_2) = (I - I_g)S_3 \dots\dots\dots (III)$$

$$I_g = 1\text{mA} = 10^{-3}\text{A} \text{ and } G = 10\Omega$$

$$10^{-3} \times 10 = (10^{-2} - 10^{-3})(S_1 + S_2 + S_3) \text{ [From equation (I)]}$$

$$\Rightarrow 10 = (10 - 1)(S_1 + S_2 + S_3)$$

$$\Rightarrow 10 = 9(S_1 + S_2 + S_3) \dots\dots\dots (IV)$$

$$10^{-3}(10 + S_1) = (10^{-1} - 10^{-3})(S_2 + S_3) \text{ [From equation (II)]}$$

$$\Rightarrow 10 + S_1 = (100 - 1)(S_2 + S_3)$$

$$\Rightarrow 10 + S_1 = 99(S_2 + S_3) \dots\dots\dots (V)$$

$$10^{-3}(10 + S_1 + S_2) = (1 - 10^{-3})(S_3) \text{ [From equation (III)]}$$

$$\Rightarrow 10 + S_1 + S_2 = (1000 - 1)S_3$$

$$\Rightarrow 10 + S_1 + S_2 = 999S_3 \dots\dots\dots \text{(VI)}$$

$$10 + S_1 = 99(S_3 + S_2) \text{ [From equation (V)]}$$

$$10 = 9(S_3 + S_2 + S_1) \text{ [From equation (V)]}$$

$$\frac{10}{9} = S_1 + S_2 + S_3 \text{ [From equation (IV)]}$$

$$\frac{10}{-99} + \frac{S_1}{-99} = -S_2 \pm S_3 \text{ [From equation (V)]}$$

$$\frac{10}{9} - \frac{10}{99} - \frac{S_1}{99} = S_1$$

$$\frac{110 - 10}{99} = S_1 + \frac{S_1}{99}$$

$$\frac{100}{99} = \frac{99S_1 + S_1}{99} \Rightarrow \frac{100}{99} = \frac{100}{99}S_1$$

$$S_1 = 1\Omega$$

$$\text{So } \frac{10}{9} = 1 + S_2 + S_3 \text{ [From equation (VI)]}$$

$$\text{Or } \frac{1}{9} = S_2 + S_3$$

$$S_3 = \frac{10}{999} + \frac{S_1}{999} + \frac{S_2}{999} \text{ [From equation (VI)]}$$

$$S_3 = \frac{10}{999} + \frac{1}{999} + \frac{S_2}{999} \left[\because S_1 = 1\Omega \right]$$

$$\Rightarrow S_3 = \frac{11}{999} + \frac{S_2}{999}$$

$$S_2 = \frac{1}{9} - S_3 \left[\because \frac{1}{9} = S_2 + S_3 \text{ from above} \right]$$

$$= \frac{1}{9} - \frac{11}{999} - \frac{S_2}{999} \text{ (From equation III)}$$

$$S_2 + \frac{S_2}{999} = \frac{111 - 11}{999}$$

$$\frac{1000}{999} S_2 = \frac{100}{999}$$

$$S_2 = 0.1\Omega$$

$$S_1 + S_2 + S_3 = \frac{10}{9} \text{ (From equation IV)}$$

$$\Rightarrow 0.1 + 0.1 + S_3 = \frac{10}{9}$$

$$\Rightarrow S_3 = \frac{10}{9} - \frac{11}{10} = \frac{100 - 99}{90} = \frac{1}{90}$$

Hence, $S_1 = 1\Omega$; $S_2 = 0.1\Omega$; $S_3 = 0.01\Omega$

4.29 Five long wires A, B, C, D and E, each carrying current I are arranged to form edges of a pentagonal prism as shown in Fig. 4.6. Each carries current out of the plane of paper.

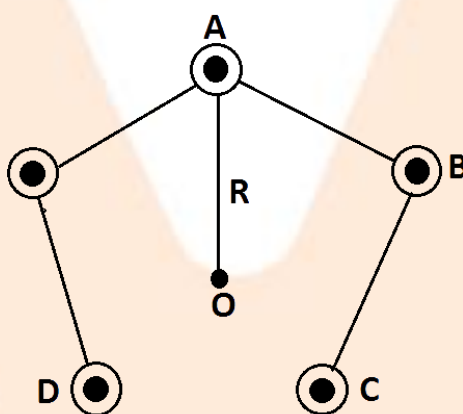
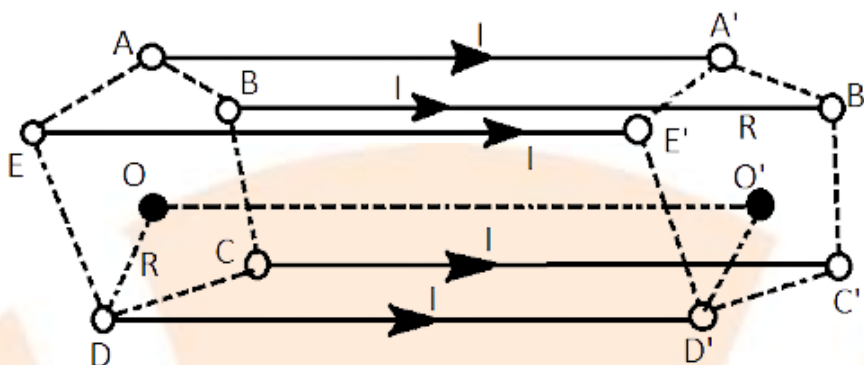


Fig. 4.6

- What will be magnetic induction at a point on the axis O? Axis is at a distance R from each wire.
- What will be the field if current in one of the wires (say A) is switched off?
- What if current in one of the wire (say) A is reversed?

Ans. (a) Figure shows that five conductors AA', BB', CC', DD' and EE' are along height of regular pentagonal prism ABCDE.



It is given that the five identical conducting wires are along the heights of regular pentagon, represented in figure above by AA', BB', CC', DD', and EE'. Axis of regular pentagon is OO' will be equidistant (R) from all five conductors, the current is passing through all five conductors are equal let (I).

As the current in all 5 conductors are equal to I and the distance of O from conductors is also equal to R then the magnitude of magnetic field due to each conductor will be equal, i.e.

$$|B_1| = |B_2| = |B_3| = |B_4| = |B_5| = B = \frac{\mu_0 I}{2\pi R}$$

The direction of magnetic field induced can be find out by right hand grip rule, then direction of induced magnetic field at 'O' due to AA' will be perpendicular to both

AA' and AO. Angles between any two consecutive magnetic field is $\frac{360^\circ}{5} = 72^\circ$

As shown in figures given below

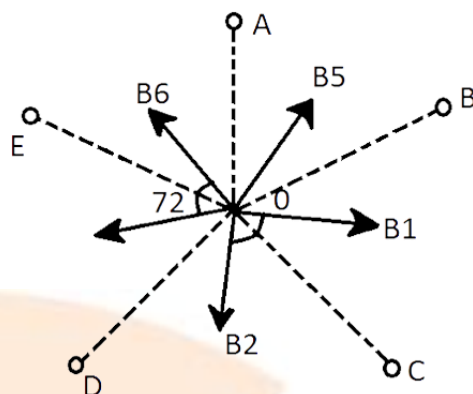
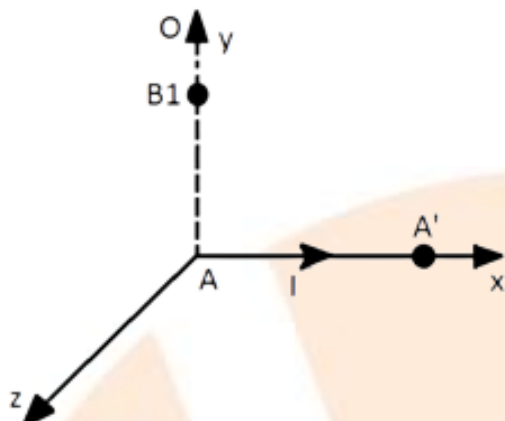


Fig. shows the direction of B_1, B_2, B_3, B_4, B_5 due to conductors AA', BB', CC', DD' and EE' respectively

As, $B = B_1 = B_2 = B_3 = B_4 = B_5$ and the angle between consecutive magnetic fields is 72° or symmetric in 360° so their resultant at O will be zero, i.e.

$$\vec{B}_1 + \vec{B}_2 + \vec{B}_3 + \vec{B}_4 + \vec{B}_5 = \mathbf{0}$$

Hence, the induced magnetic induction at O due to the five conductors as shown in figure is zero.

(b) When current in AA' is switched off, then $B_1 = \mathbf{0}$ and resultant becomes

$$R = B_2 + B_3 + B_4 + B_5$$

But from (a) part, $B_1 + B_2 + B_3 + B_4 + B_5 = \mathbf{0}$

$$\text{Or } \vec{B}_2 + \vec{B}_3 + \vec{B}_4 + \vec{B}_5 = -\vec{B}_1$$

$$R = -B_1$$

$$R = \frac{\mu_0 I}{2\pi r}$$

i.e. direction of resultant is opposite to $\vec{B}_1$.

(c) Here, on reversing the current in AA' , direction of magnetic field due to AA' becomes $-\vec{B}_1$,

$$R = B_2 + B_3 + B_4 + B_5$$

$$\therefore |-\vec{B}_1| = |B_2| = |B_3| = |B_4| = |B_5| = B$$

$\therefore$ Net induced magnetic field at O becomes

$$-B + B + B + B + B = 3B$$

$$R = \frac{3\mu_0 I}{2\pi r}$$

